Template-matching recovery of pile-up in the ZGZ RUN1 records

Method, pulse tail and undershoot, and first results

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October 2026

Summary

We recover the pulses that pile up in the triggered records of the ZGZ RUN1 acquisition with template matching (TM), so that every pulse, alone or overlapping, has a time and an energy. Templates and fit settings are learned from the clean pulses of each channel. The key difficulty is that the acquisition stops every record before the pulse has returned to the baseline, so the long tail and the undershoot below the baseline, which add up in pile-up, are never seen in a single clean record; we measure them directly from the baseline of the following record. On channel 1, a second pulse above half the photopeak is found in 98 % of the cases when it arrives 4–10 µs after the first, with an energy error of about 1–2 %, and the pulses recovered from pile-up have the same photopeak as the clean ones. A self-contained Python package runs each channel with one command and writes the time and energy of every pulse. A second, faster-decaying pulse shape found in the data is flagged below.

Important: two pulse shapes. Besides the main pulse shape, the data contain a second population of single pulses that decay faster (34 %, 3 % and 25 % of the clean pulses of channels 0, 1 and 2). Both are currently described by one set of templates per channel, so the templates are a mixture of the two shapes, weighted by their abundance at each amplitude. This has to be either solved, with a template set per shape, or accepted as a known approximation; see Section 3.4.

1. Data and pipeline

Each channel is a text file, `c<ch>.txt`, with one triggered record per block and one sample per line (time within the record, height, and the output of the acquisition's VST filter), sampled every 128 ns with an 8-bit ADC. A second file, `out.txt`, gives the time of every record since the start of the run, which is not in `c<ch>.txt`, together with the maximum of the VST filter, which we use to match records and rows (99.995 % of the records of channel 1 matched). A clean record has about 20 samples of baseline before the pulse and ends some 40–60 µs later, earlier for small pulses; a pile-up record is extended by every new pulse, up to 167 µs. The records come in bursts: although the mean rate of channel 1 is 0.18 records per second, about 6 % of them start less than 400 µs after the previous one. Fig. 1 shows the steps of the analysis.

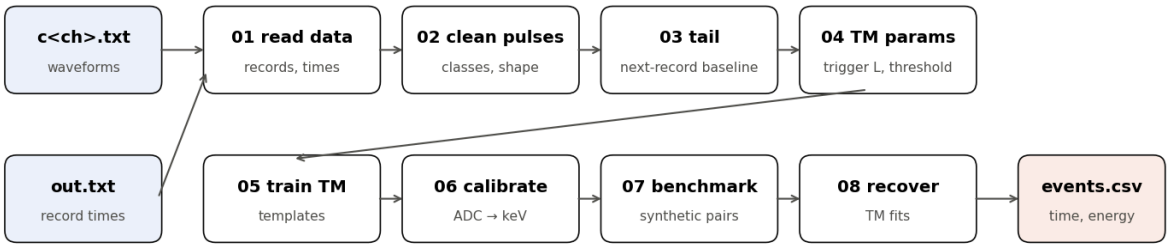


Figure 1. Steps of the analysis, one script each. The clean pulses of a channel (from all runs, if several are analysed together) give the pulse shape, tail and templates, which are then used to fit every run. The record times from `out.txt` are used for the pulse times and for the measurement of the pulse tail.

2. Method

2.1. Clean pulses

The clean pulses, from which the pulse shape and the templates are learned, are selected with the area–amplitude criterion we use for $LaBr_3$(Ce), completed with checks that slow pulses need (Fig. 2). Single pulses lie on one curve of area against amplitude, slightly concave here because records end earlier for small pulses; we follow it outward from the photopeak, fit it with a quadratic, and reject records whose area exceeds it by more than three robust standard deviations, since pile-up adds area. Records below the curve are kept: they are single pulses of a second, faster-decaying shape (Section 3.4). This band does not remove two pulses that arrive within a few microseconds of each other: with a 5 µs flat top and a 50 µs decay their areas and amplitudes simply add up, and above the photopeak, where true single pulses are rare, most records inside the band would be pile-up. We therefore also require exactly one trigger of the TM fast filter, Eq. (3), with a short window ($L = 4$ samples) so that it re-arms soon after the pulse's own rise, and a threshold of 5.5 times its noise (3.1 ADC on channel 1); a 10–90 % rise of at most 20 samples, which catches pile-up within the rise; a flat pre-trigger baseline, which rejects records that start on the tail of an earlier pulse; and no saturated sample. Table 1 lists the outcome; the records above the band are mostly small pile-up below the trigger threshold.

Table 1. Classes of the 2,036,169 records of channel 1 (first rule that applies). All except the clean and saturated ones are fitted by TM.

Class	Records	Fraction
clean	1,792,079	88.0 %
on an earlier tail (baseline)	129,589	6.4 %
saturated	5,402	0.3 %
no pulse	367	0.0 %
pile-up: second trigger	77,648	3.8 %
pile-up: slow rise	30,451	1.5 %
above the area band	633	0.0 %

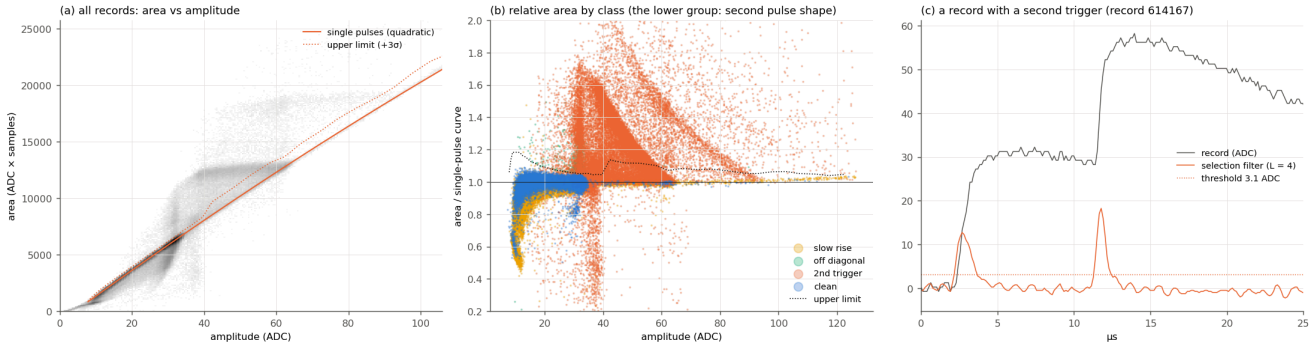


Figure 2. Selection of the clean pulses of channel 1. (a) Area against amplitude of all records, with the single-pulse curve and its upper limit. (b) Area relative to the curve for each class; clean pulses below 1 belong to the second pulse shape. (c) A record with a second trigger: the selection filter (orange) crosses the threshold once on each rise.

2.2. Pulse shape and amplitude

The flat top is sampled in steps of one ADC count, so the maximum sample falls at a random place on it. We align pulses on their 50 % rise crossing, t_{50}, and take as reference point the centre of the flat top, $D = 27$ samples later. The mean shape T is the median of the aligned clean pulses divided by their amplitude, and the amplitude of a clean pulse is the least-squares scale of that shape over its recorded samples,

$$A = \frac{\sum_j x_j \, T(j - t_{50})}{\sum_j T(j - t_{50})^2}. \tag{1}$$

The pulse rises from 10 to 90 % in 1.4 µs and the clean pulses have a very uniform shape (Fig. 3).

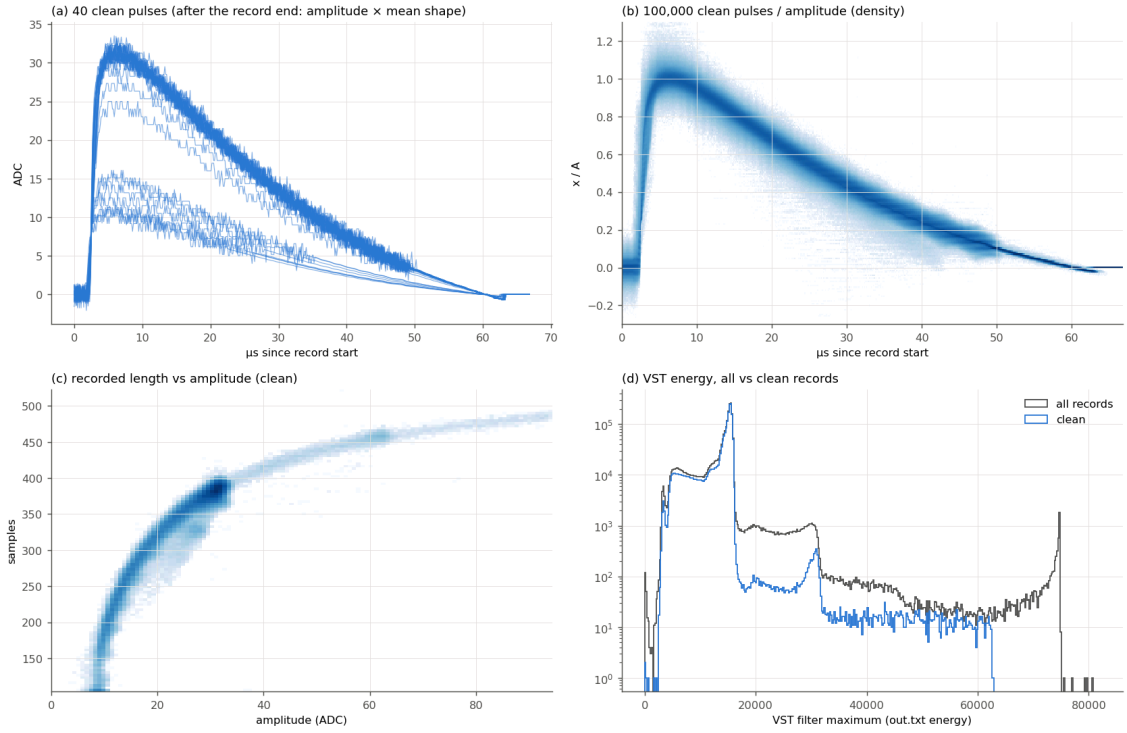


Figure 3. Clean pulses of channel 1. (a) Examples, continued after the end of each record with their amplitude times the mean shape. (b) Density of 100,000 clean pulses divided by their amplitude. (c) Record length against amplitude. (d) VST energy of all and of the clean records.

2.3. The tail and the undershoot

The records end too early. A photopeak pulse is still 3–4 ADC above the baseline when its record stops, and only pulses above about 70 ADC reach the zero crossing within their record (Fig. 4). A template built from clean records therefore knows nothing about the end of the pulse, and in a crowded record the missing tails accumulate: Fig. 5(a) shows a record with eleven pulses where the data fall 20 ADC below the fit after 60 µs. The data show what is missing: after crossing zero the pulse goes below the baseline, an undershoot.

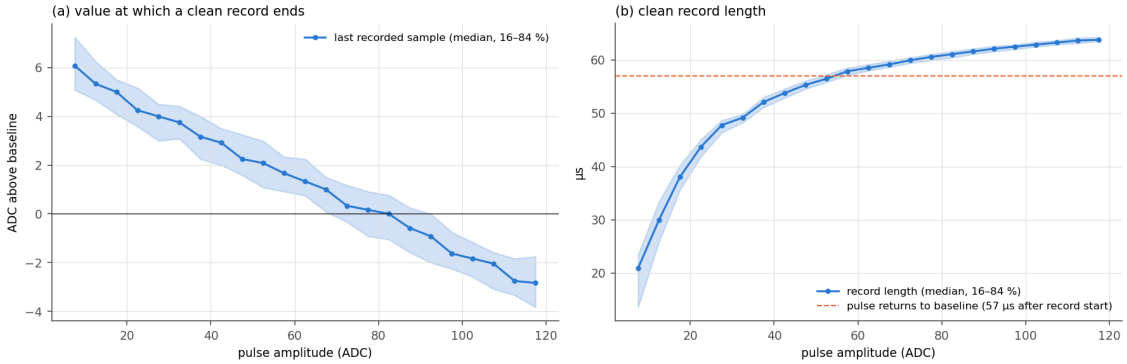


Figure 4. (a) Height of the last sample of the clean records of channel 1 against the pulse amplitude. (b) Record length against amplitude, compared with the time at which the pulse actually returns to the baseline (dashed).

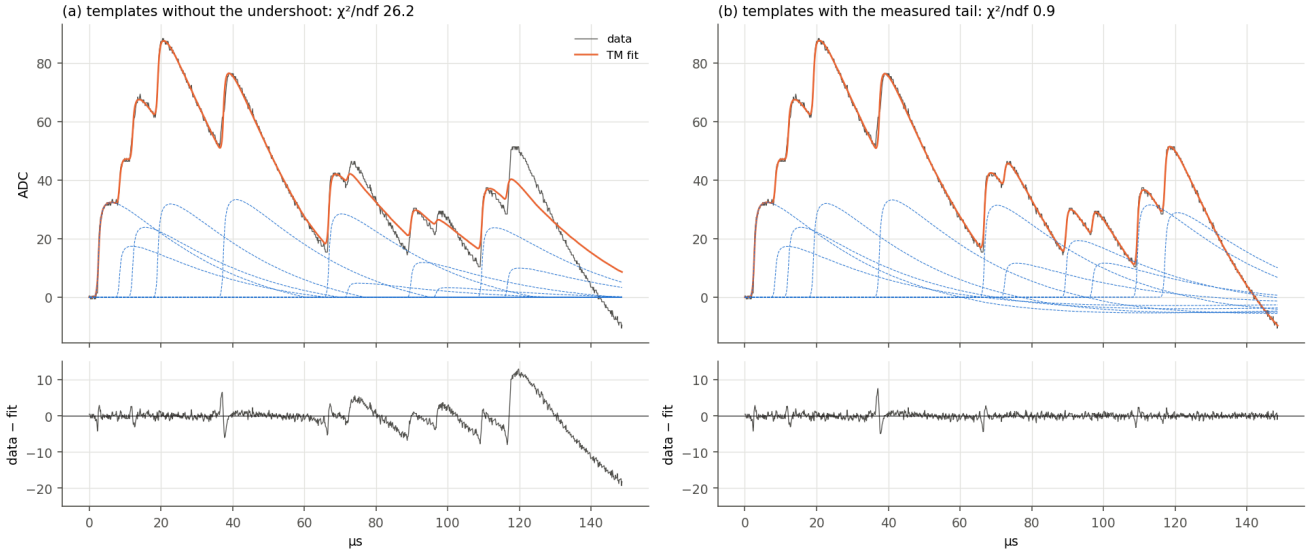


Figure 5. A record with eleven pulses fitted by TM (orange; single pulses dashed) (a) with templates that stop at the zero crossing and (b) with the measured tail; lower panels: residual.

Direct measurement. When a record starts while the previous pulse is still in its tail, its pre-trigger baseline is the value of that tail. Taking pairs of consecutive records in which the first is a single clean pulse of amplitude A_{prev}, with nothing before it for at least 1 ms, each pair gives one point of the tail of an undisturbed pulse,

$$y = \frac{b_i - b_0}{A_{\mathrm{prev}}}, \qquad k = \frac{t_i - t_{\mathrm{prev}}}{\Delta t} + \frac{N_b - 1}{2} - (t_{50,\mathrm{prev}} + D), \tag{2}$$

where b_i is the baseline of the new record (mean of its $N_b = 12$ first samples), b_0 that of isolated records, t_i and t_{prev} the record times, Δt the sampling period and k the time after the previous pulse's reference point. Channel 1 gives 67,631 pairs (Fig. 6): the pulse

crosses zero 54.8 µs after its reference point, reaches -0.166 times its amplitude at 107 µs and recovers slowly. The template uses this curve after the end of the clean-pulse shape.

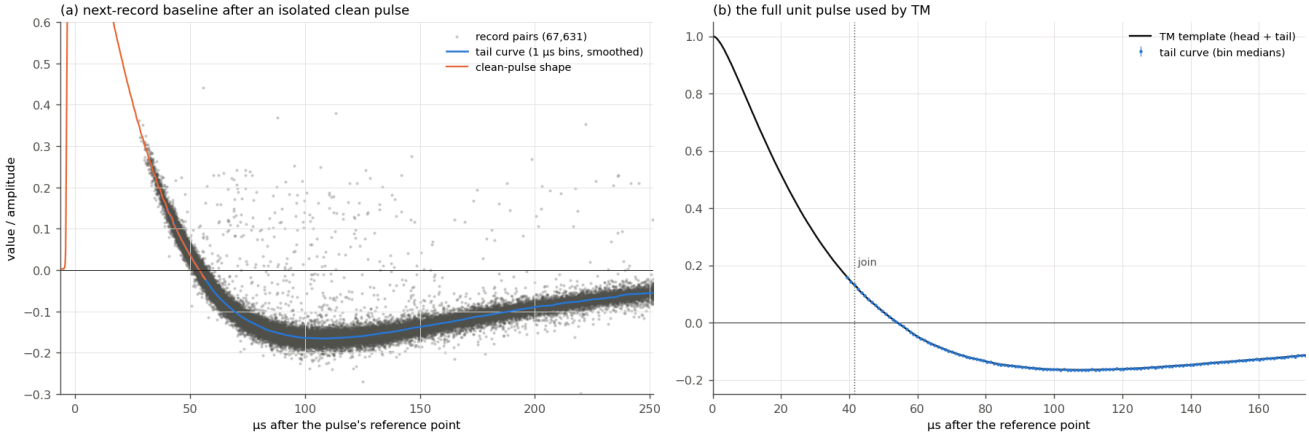


Figure 6. (a) Baseline of the next record divided by the amplitude of an isolated clean pulse, against the time since that pulse (grey), with the binned and smoothed curve (blue) and the clean-pulse shape (orange). (b) The unit pulse used by TM: clean-pulse shape, then the measured tail.

Cross-checks. Two indirect estimates of the tail, from the residual of fitted pile-up records and from saturated single pulses, agree with it up to about 75 µs (the saturated pulses end at 62 µs, and beyond 75 µs the pile-up estimate, constrained by few records, is shallower), and a fit of an AC-coupled pulse, $y \propto (1 - t/\tau)\, e^{-t/\tau}$ with $\tau = 55$ µs, predicts the depth and position of the minimum (Fig. 7). On long pile-up records, tails that return to the baseline faster, or a constant undershoot, fit clearly worse.

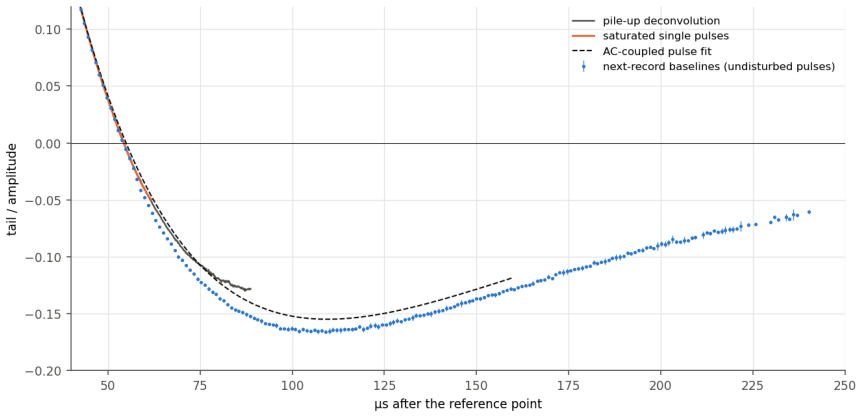


Figure 7. Tail per unit amplitude: direct measurement (blue) against the indirect estimates from pile-up records (grey) and saturated pulses (orange) and the AC-coupled pulse (dashed).

Linearity. TM assumes that pulses add up. In 11,424 two-pulse records with the second pulse 20–40 µs after the first, before the first pulse's undershoot has started, we fitted a free scale s of that undershoot: the real records give $s = 0.88$–0.92, growing with the delay, while synthetic records built to add up exactly give 0.987 (and 0.892 when built with 10 % less undershoot), with no dependence on the delay (Fig. 8). The undershoot is therefore present but about 10 % smaller when another pulse follows soon, a weak non-linearity of the electronics worth about 0.5 ADC for a photopeak pulse, below the noise of a single sample. The template keeps the undisturbed tail, which fits the crowded records best (Fig. 8a).

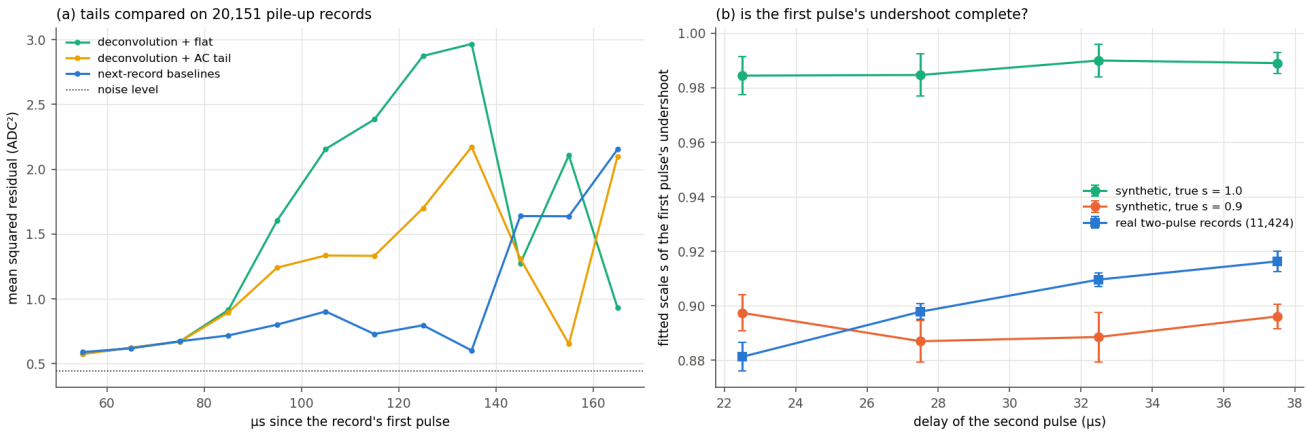


Figure 8. (a) Residual of 20,151 long pile-up records with three tails, against the time since the first pulse. (b) Fitted scale of the first pulse's undershoot in two-pulse records: real records (blue) and synthetic linear records with true $s = 1$ and 0.9.

2.4. Template matching

The fitter is the template-matching code of our $LaBr_3$(Ce) work. A record is a sum of pulses,

$$x(t) = \sum_p A_p \, h_{t-t_p}(A_p), \qquad h_k(A) = a_{0,k} + a_{1,k}A + a_{2,k}A^2, \tag{3}$$

where t_p is the reference point of pulse p and h_k the normalised template at offset k, quadratic in the amplitude; the coefficients are fitted offset by offset over 250,000 clean pulses up to the start of the measured tail, and from there h_k is the tail itself, out to the end of the longest record. Pulses are found with the fast filter

$$f_i = \frac{1}{L}\sum_{m=0}^{L-1} x_{i+m} - \frac{1}{L}\sum_{m=1}^{L} x_{i-m}, \tag{4}$$

one pulse is placed per trigger, and further pulses are added where the residual is large if χ^2/ndf improves enough. All settings are measured on the clean pulses (Fig. 9): the filter length is limited to $L = 10$ (1.28 µs) by the short pre-trigger, the threshold, 2.27 ADC, is the lowest at which at most 0.2 % of the clean pulses trigger twice, and pulses down to about 3 ADC trigger. Triggers in the last L samples of a record are ignored, since there the filter's look-ahead window runs past the end and a tail below the baseline looks like a rising edge. On single pulses the TM amplitude equals Eq. (1) to 0.01 %.

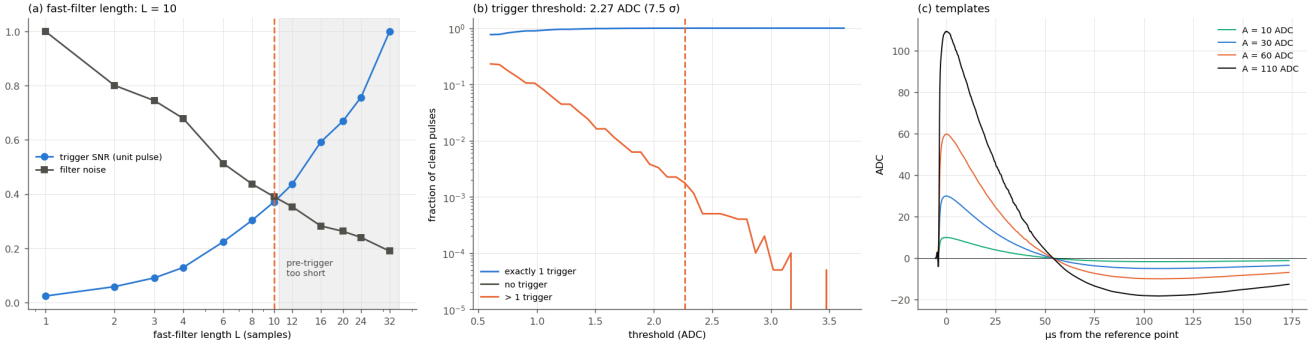


Figure 9. TM settings of channel 1: (a) trigger signal-to-noise against the filter length, (b) clean pulses with zero, one or more triggers against the threshold, (c) templates at four amplitudes, with the undershoot.

2.5. Energy and benchmark

The amplitude is converted to energy with $E = c\,A$, placing the photopeak of the clean pulses at its energy (766 keV for this run): $c = 24.54$ keV/ADC on channel 1, with 6.0 % FWHM. The sum peak then falls at 1.976 times the photopeak energy (Fig. 10), so a proportional calibration describes both peaks to within 1.2 %. To measure the performance we build pile-up with a known answer from the last 10 % of the clean pulses, which are kept out of the templates: pairs of pulses are summed with delays between 0.2 and 100 µs, each pulse continued after its record with the measured unit pulse plus noise, and fitted like the real records. A pulse is found if a fitted pulse lies within 0.5 µs and 10 % of it.

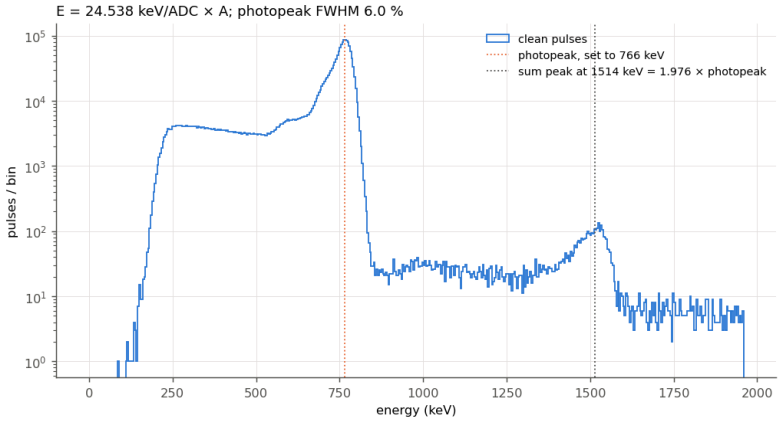


Figure 10. Calibrated spectrum of the clean pulses of channel 1, with the photopeak, which fixes the calibration, and the sum peak, which checks it.

3. Results

3.1. Benchmark

Both pulses are found in more than 95 % of the pairs from about 4 µs on, with an energy error of 1–3 % that falls to about 1 % beyond 10 µs (Fig. 11, Table 2); below 1 µs the pulses are not separated. The losses at intermediate delays are small second pulses: for second pulses above half the photopeak the second pulse is found in 97 % of the pairs at 2.5–4 µs and 98 % at 4–10 µs, while for smaller ones 72 % and 95 %, since a small pulse on the end of a larger pulse's rise does not trigger the filter on its own. The time error, 60–70 ns, is dominated by the reference used (the t_{50} of each pulse, rounded to the 128 ns grid).

Table 2. Benchmark on 5,000 synthetic pairs of channel 1: pairs with both pulses found, and robust standard deviation of the energy and time errors (first / second pulse).

Delay (µs)	Pairs	Both found	Energy error (%)	Time error (ns)
0.32	737	2.2 %	16.9 / 30.2	160 / 289
0.71	550	6.7 %	58.8 / 49.6	160 / 263
1.2	290	27.9 %	24.8 / 23.3	144 / 107
1.7	209	53.1 %	11.0 / 11.5	119 / 78
2.4	318	79.2 %	5.4 / 5.3	102 / 70
3.9	429	95.3 %	2.6 / 2.7	88 / 62
6.3	380	98.4 %	1.3 / 2.0	72 / 63
9.8	333	97.9 %	1.1 / 1.8	69 / 65
15	439	98.2 %	1.2 / 1.7	70 / 64
24	322	98.4 %	1.2 / 1.3	66 / 70
37	305	97.7 %	1.3 / 0.7	67 / 67
56	367	98.6 %	1.1 / 0.3	65 / 70
84	321	99.4 %	1.3 / 0.3	62 / 77

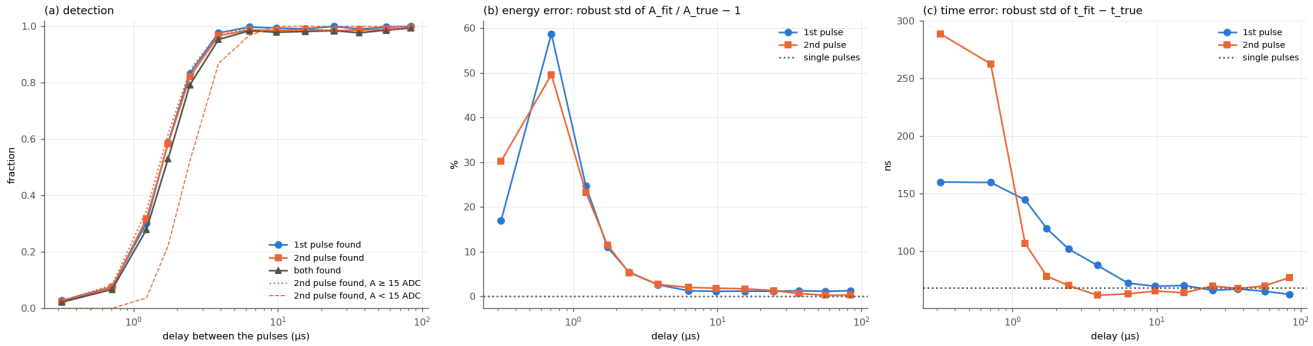


Figure 11. Benchmark against the delay between the two pulses: (a) fraction found, also split by the amplitude of the second pulse; (b) energy error; (c) time error; dotted: single pulses.

3.2. Real pile-up

TM fitted the 238,321 records of channel 1 that are not clean, in 6 minutes with three processes. 82,554 records are resolved into two or more pulses, giving 178,162 pulses, and the remaining fitted records hold a single pulse (mostly the slow-rise and off-band classes, which are single pulses of a slightly different shape). With the 1,792,079 clean pulses and those on an earlier tail, the table of channel 1 holds 2,121,740 pulses. The fits follow the data also in the most crowded records (Figs. 12, 13). Pulses resolved from pile-up have their photopeak at 768 keV with 5.9 % FWHM, against 766 keV and 6.0 % for clean pulses (Fig. 14). A few fitted pulses are flagged and left out (20,488 outside their record, mostly the earlier pulse whose tail a record starts on), and pairs closer than 1 µs, which the benchmark shows cannot be separated, should be treated as one pulse. Resolved pulses are spaced from a few µs to about 50 µs, beyond which a later pulse starts its own record (Fig. 15).

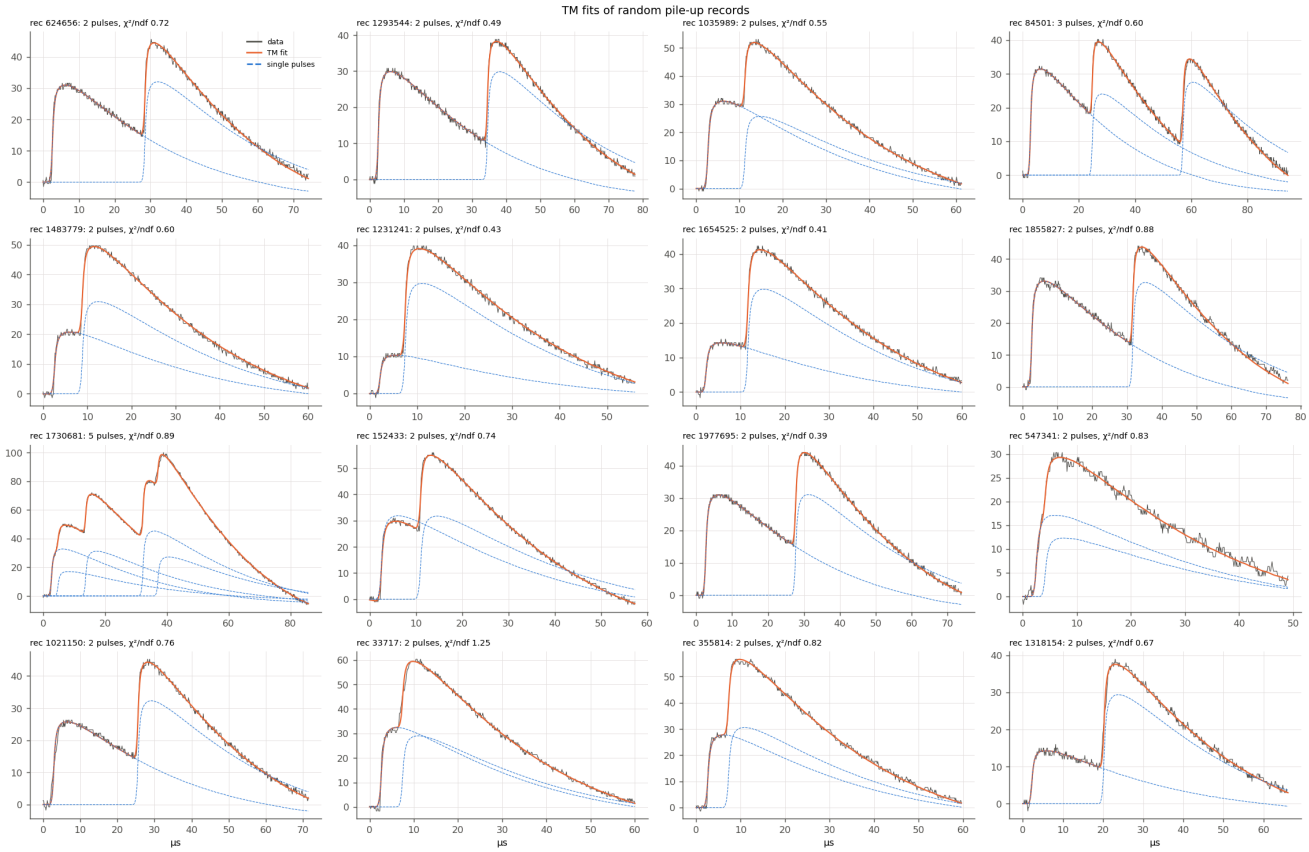


Figure 12. TM fits of random pile-up records of channel 1: data (grey), fit (orange) and the individual pulses (dashed).

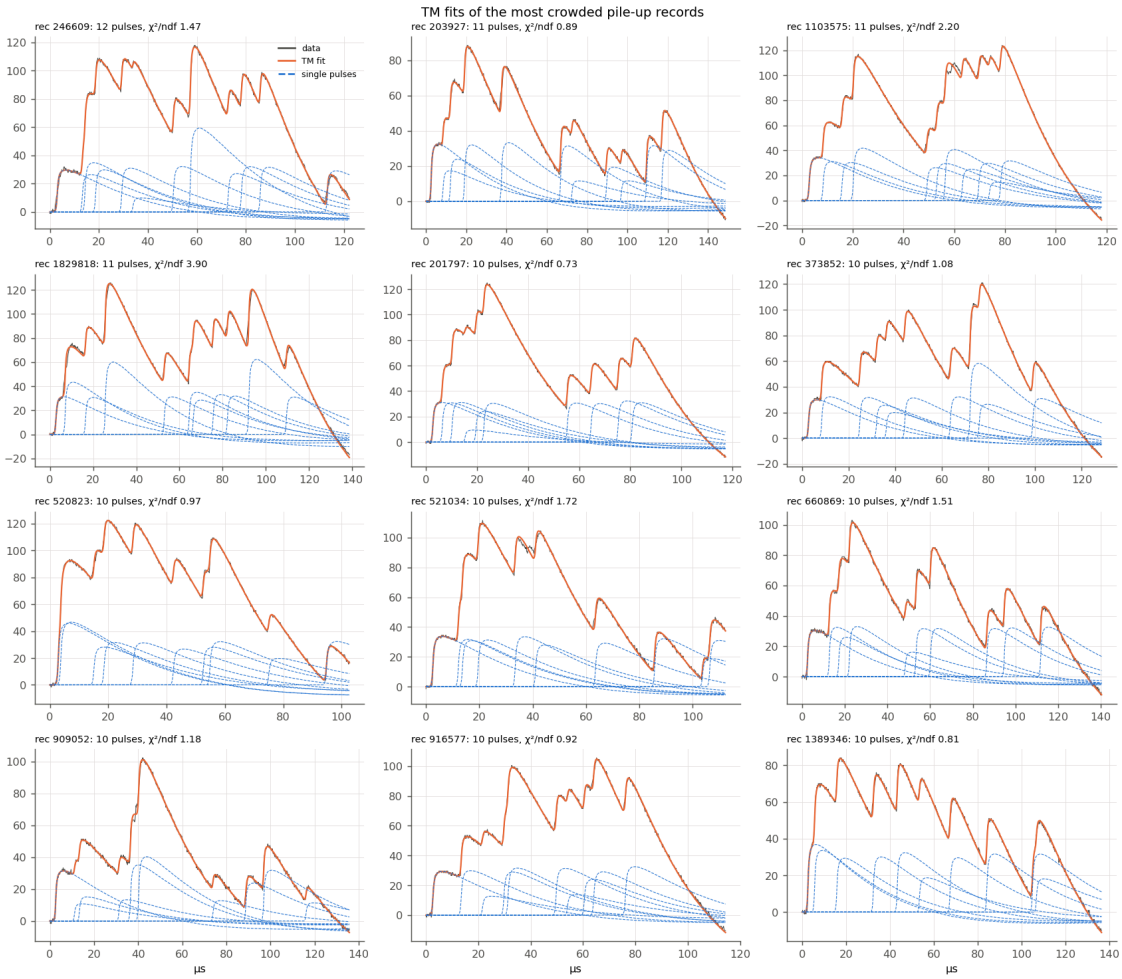


Figure 13. TM fits of the most crowded pile-up records of channel 1.

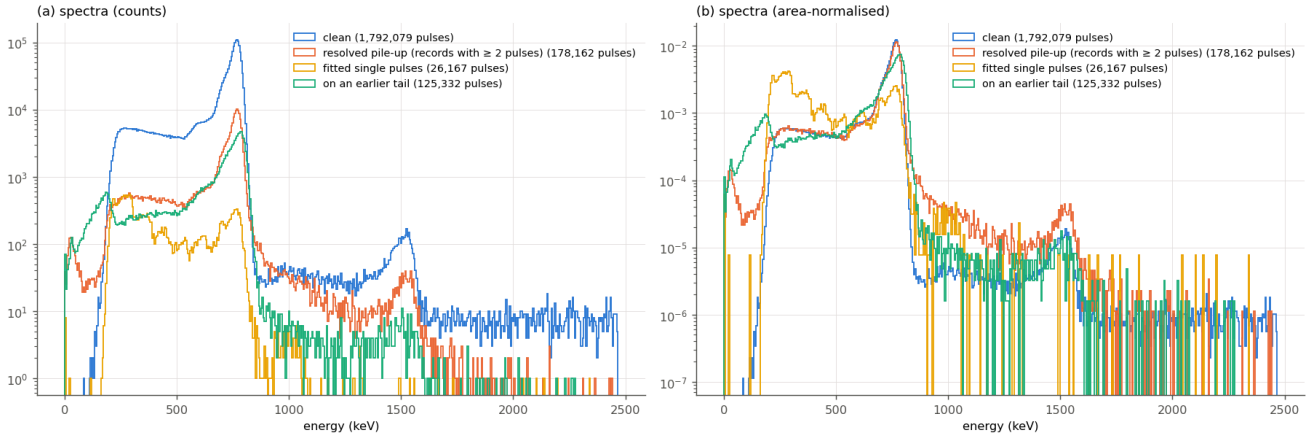


Figure 14. Energy spectra of channel 1: clean pulses, pulses resolved in pile-up records, single pulses fitted by TM, and pulses in records that start on an earlier tail; (a) counts, (b) normalised.

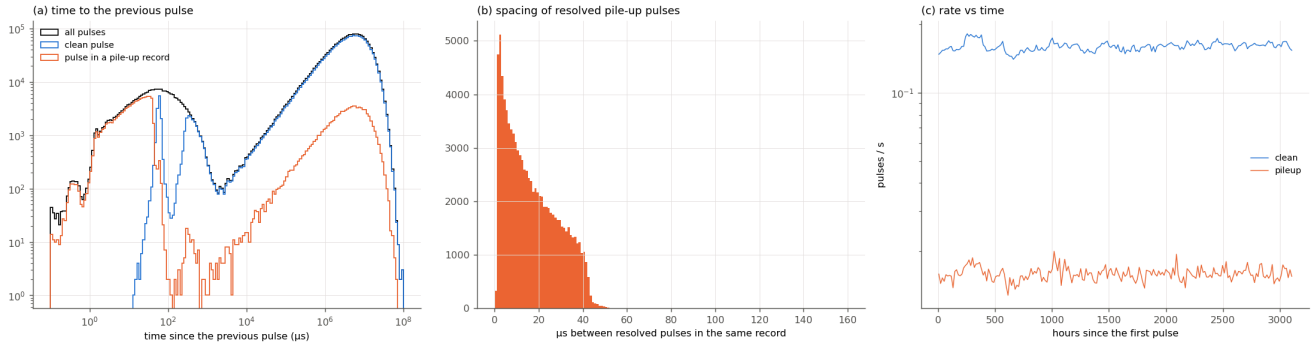


Figure 15. Time distributions of channel 1: (a) time to the previous pulse; (b) spacing of the pulses resolved in the same record; (c) rate during the run.

3.3. All channels

The three channels were analysed independently with the same procedure (Table 3). Their ADC steps differ slightly, but the pulse shape, the tail and undershoot and the TM settings are very similar, so the method transfers between channels without hand tuning. The benchmark differs only through the small second pulses: channels 0 and 2 have more small pulses, and therefore a lower overall efficiency at short delays. The longest record of channel 0 (350 µs) exceeds the range of its measured tail, which the package then continues with the exponential decay of its last 50 µs.

Table 3. Main numbers of the three channels, each analysed independently.

	ch 0	ch 1	ch 2
records	2,611,405	2,036,169	2,469,207
matched to `out.txt`	99.997 %	99.995 %	99.996 %
clean	92.2 %	88.0 %	92.4 %
pile-up (trigger, rise)	3.5 %	5.3 %	3.4 %
above the area band	0.0 %	0.0 %	0.0 %
second pulse shape (of clean)	34.3 %	2.6 %	24.7 %
on an earlier tail	3.7 %	6.4 %	3.8 %
baseline noise (ADC)	0.55	0.60	0.50
tail: zero crossing (µs)	52.7	54.8	54.8
tail: minimum (× amplitude)	-0.174	-0.166	-0.164
TM threshold (ADC)	1.76	2.27	1.53
photopeak FWHM	6.1 %	6.0 %	5.6 %
2nd pulse found, 4–10 µs, A ≥ ½ peak	98.3 %	98.4 %	99.2 %
2nd pulse found, 4–10 µs, A < ½ peak	72.4 %	95.4 %	79.5 %
pulses resolved in pile-up records	139,192	178,162	157,455

3.4. Two pulse shapes

The relative area of the clean pulses, their area divided by the single-pulse curve, shows two groups in every channel (Fig. 16a): the main shape at 1 and a second one at 0.6–0.85, depending on the channel, which contains 34 %, 3 % and 25 % of the clean pulses of channels 0, 1 and 2. The second shape rises like the main one but decays faster (Fig. 16c),

and its spectrum is different, dominated by small pulses below about 400 keV (Fig. 16b). Two well-separated decay times are what pulse-shape discrimination between particle types looks like, so this population may be of physical interest; its origin should be checked with the knowledge of the detectors.

The package currently keeps both shapes together: the clean set, and therefore the templates and the measured tail, are a mixture of the two, weighted by their abundance at each amplitude. Since the second shape is concentrated at small amplitudes, the mixture is close to the main shape for photopeak pulses but not for small ones, and TM fits pulses of either shape with this compromise. The amplitude of a second-shape pulse, and with it its energy, is therefore biased, and in pile-up its faster decay leaves residuals that the fit can absorb with small extra pulses. There are two ways forward: to accept this approximation, which mainly affects the low-energy part of the spectra and is small for channel 1, or to solve it by building a template set for each shape (both shapes go through the same steps of the package) and fitting each pulse with the set that describes it best. The `relative_area` column of the result table already separates the two shapes for clean pulses.

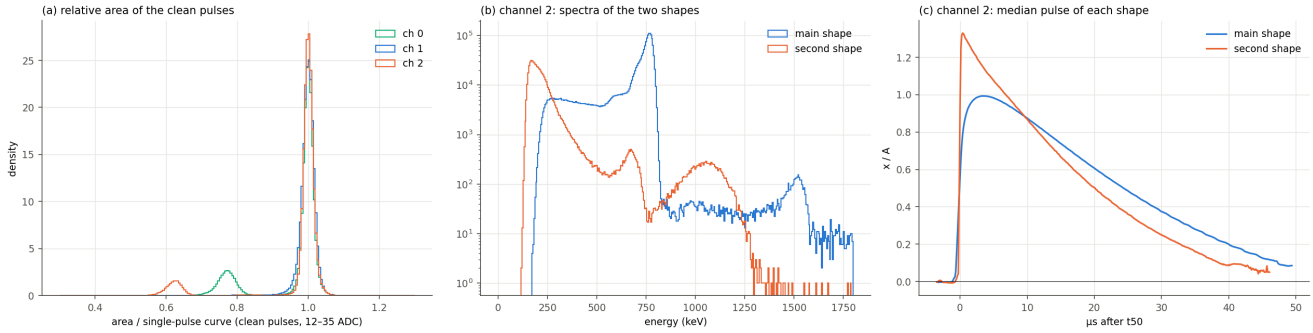


Figure 16. Two pulse shapes. (a) Area of the clean pulses relative to the single-pulse curve, 12–35 ADC, per channel. (b) Channel 2: energy spectra of the clean pulses of each shape. (c) Channel 2: median pulse of each shape, normalised to its amplitude.

4. Open points

The most important open point is the second pulse shape of Section 3.4, which should be either solved with a template set per shape or accepted as an approximation once its origin is known. Several other points could not be settled from the data alone. Firstly, the rule that ends a record: the last sample decreases with the amplitude, which suggests a fixed time after the trigger or a condition on the VST filter. Secondly, the weak non-linearity of the undershoot points to an active element such as a baseline restorer, whose time constants would tell us whether the template should depend on the time since the previous pulse. Thirdly, records that start on an earlier tail are fitted with a constant baseline, which is only approximate while that tail is still changing. Finally, saturated pulses keep their time only, and the calibration is a single constant that known lines can replace. In further works we plan to add our neural-network pile-up method to the package; an optional step already exports the clean pulses completed with the measured tail, which is what is needed to build its training data.

5. Conclusions

The results presented in this work show that the pile-up of the ZGZ RUN1 records can be resolved with template matching once the pulse tail, which the acquisition never records for a single pulse, is measured from the baseline of the following records. Every setting is measured from the data of each channel, resolved and clean pulses share the same energy scale, and a self-contained package writes the time and energy of every pulse.